

PharmaCare Admin Dashboard

Complete System Documentation & Flow Analysis

Comprehensive presentation covering Use Case Diagrams, Class Diagrams,
System Architecture, Process Flows, and Strategic Recommendations

Pharmacy Management System

PREETHIKA P

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Executive Summary

01

System Overview

The **PharmaCare Admin Dashboard** is a centralized command center for Super Administrators to manage the entire pharmacy ecosystem. It provides comprehensive oversight of clinics, users, inventory, and system health.

Key Capabilities

Module	Description	Key Features
Dashboard Analytics	Real-time KPIs and metrics	Revenue, Orders, Inventory, Users
Clinic Management	Full lifecycle management	Enrollment, Verification, Suspension
User Management	Role-based access control	Admin, Pharmacist, Customer roles
Inventory Oversight	Global stock monitoring	Low stock alerts, Reorder triggers
Audit & Compliance	Activity logging	Action tracking, IP logging

Document Contents

This presentation includes 8 professional UML diagrams, detailed process flows, data model documentation, API endpoints, and strategic recommendations.

System Architecture Diagram

Three-Tier Architecture Overview

The system follows a modern three-tier architecture pattern ensuring separation of concerns, scalability, and maintainability.

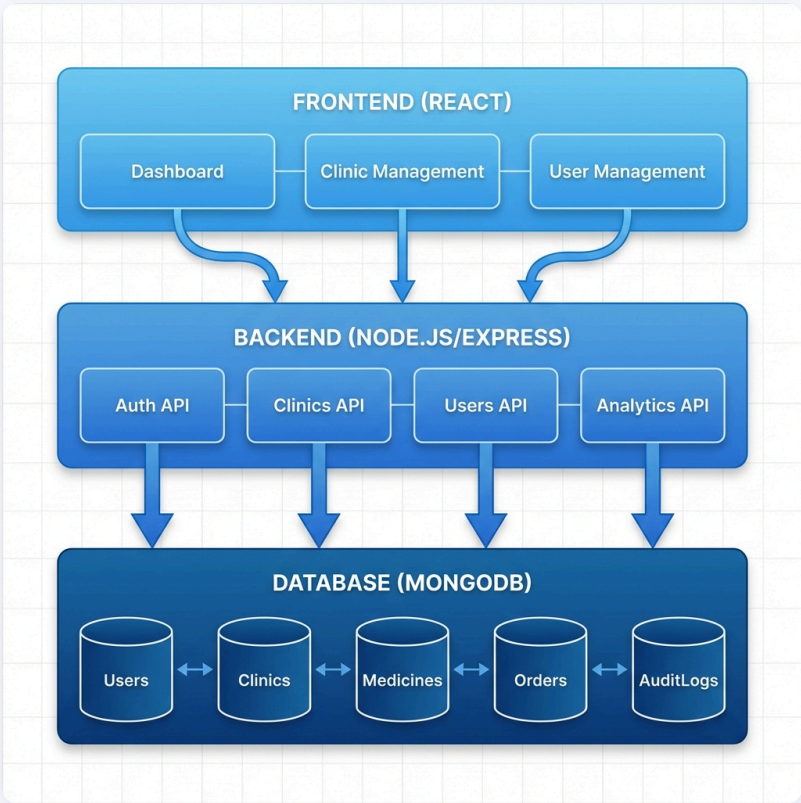


Figure 1: Three-Tier System Architecture

Layer Details

Layer	Technology	Components
Frontend	React.js	Dashboard, Clinic Management, User Management
Backend	Node.js + Express	Auth API, Clinics API, Users API, Analytics API
Database	MongoDB	Users, Clinics, Medicines, Orders, AuditLogs

Use Case Diagram

03

Actor Interactions & System Boundaries

This UML Use Case Diagram illustrates all functional interactions between administrators and the system.

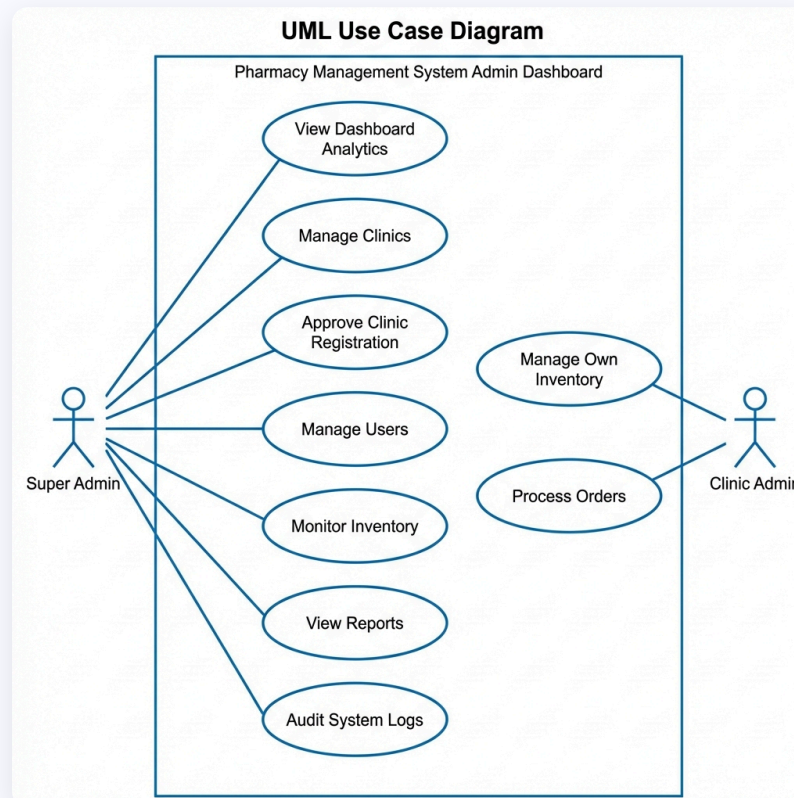


Figure 2: UML Use Case Diagram - Admin Dashboard

Actor Definitions

Actor	Role
Super Admin	Full system access
Clinic Admin	Clinic-scoped access

Primary Use Cases

- View Dashboard Analytics
- Manage Clinics
- Approve Clinic Registration
- Manage Users
- Monitor Inventory
- Audit System Logs

Class Diagram

04

Data Model Structure & Relationships

The UML Class Diagram shows the core entities and their relationships in the MongoDB database.

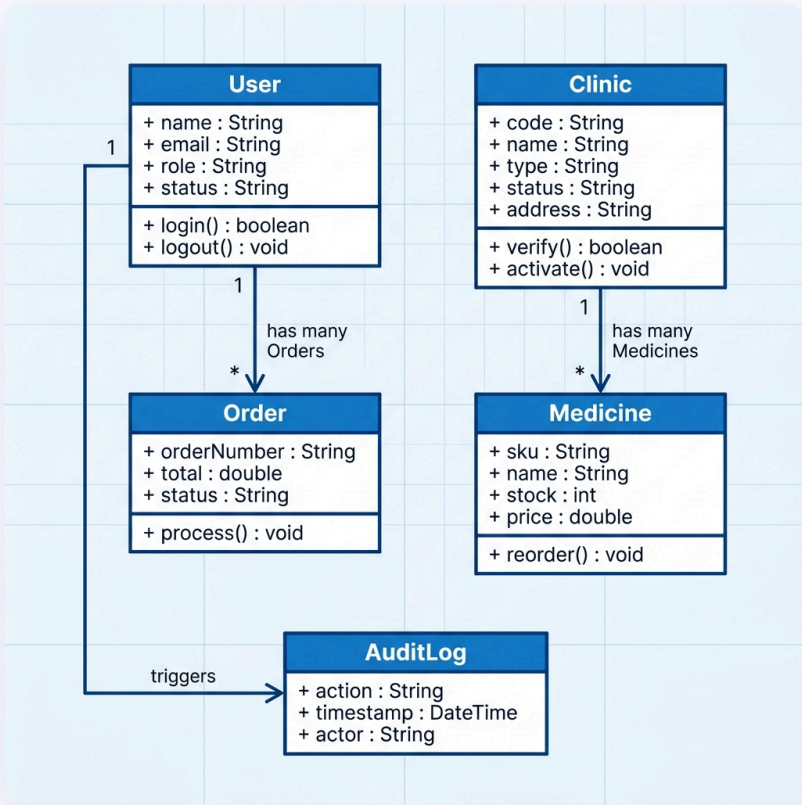


Figure 3: UML Class Diagram - Data Model

Entity Specifications

User Entity

name String

email String (unique)

role Enum

status String

Clinic Entity

code String (auto-gen)

name String

type Enum

status String

Relationships: User → Order (1:N), Clinic → Medicine (1:N), User → AuditLog (1:N)

Clinic Enrollment Process Flow

05

Step-by-Step Enrollment Workflow

This flowchart details the complete process of adding a new clinic to the platform.

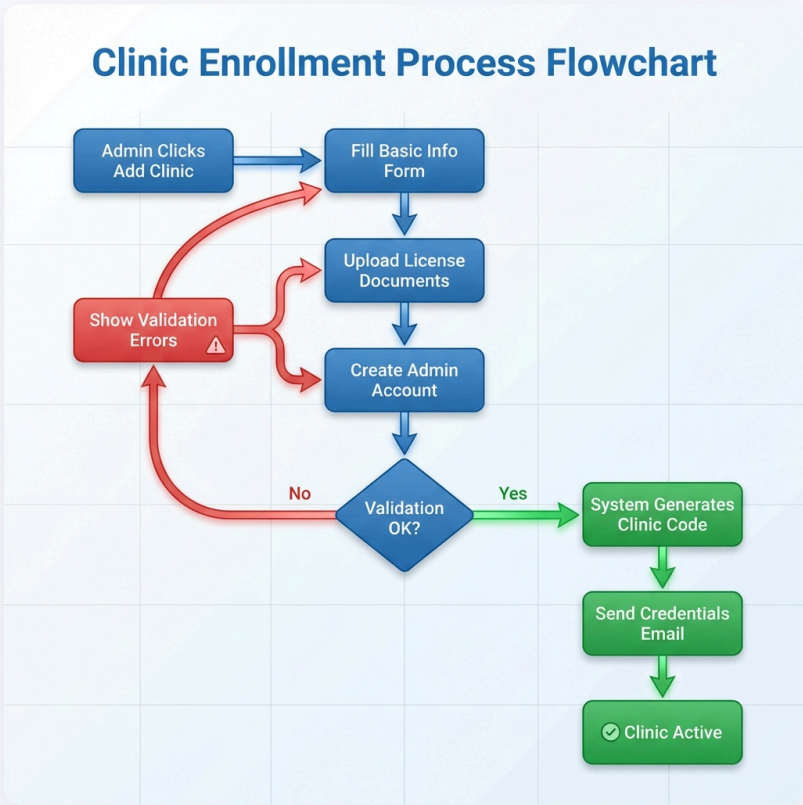


Figure 4: Clinic Enrollment Process Flowchart

Process Steps

Step	Action	System Response
1	Admin clicks "Add Clinic"	Open enrollment form
2	Fill Basic Information	Validate required fields
3	Upload License Documents	Store in filesystem
4	Create Admin Account	Generate temporary password

Step	Action	System Response
5	Validation Check	Server-side schema validation
6	On Success	Generate clinic code
7	Email Notification	Send credentials to clinic admin

User Management Process Flow

Complete User Lifecycle Management

This diagram shows all user management operations available to the Super Admin.

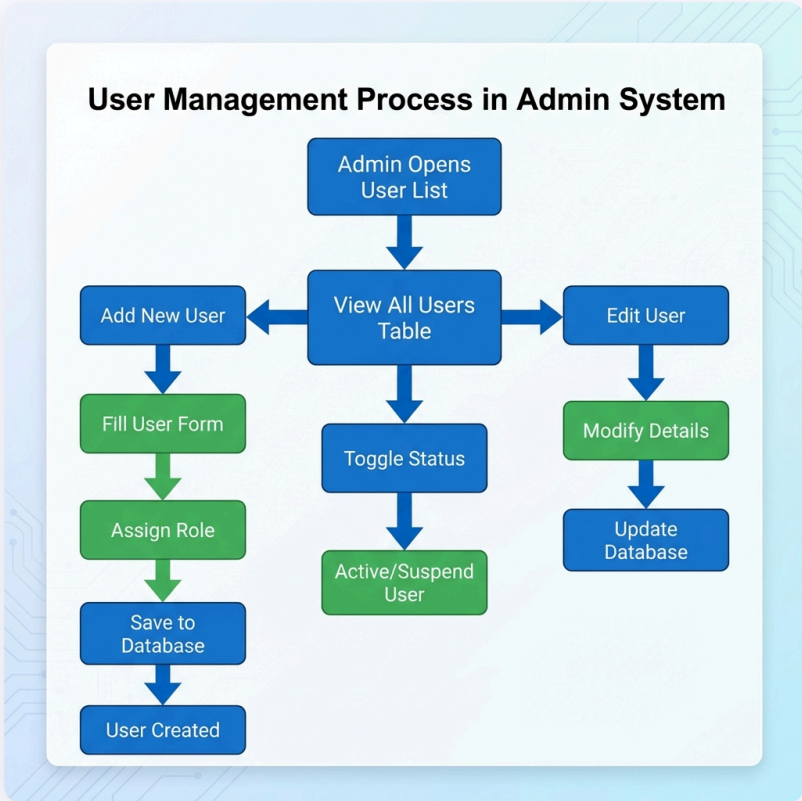


Figure 5: User Management Process Flowchart

Available Operations

Operation	Description	Result
View Users	Display all users in table	Filterable by role/status
Add User	Create new user account	Assign role, send email
Edit User	Modify user details	Update database
Toggle Status	Activate/Suspend user	Immediate access control

Role Hierarchy

Super Admin (Full Access) |— Clinic Admin (Clinic-Scoped Access) | — Pharmacist
(Inventory + Orders) — Customer (Order Placement Only)

Data Flow Diagram (DFD)

Complete Data Movement Across System

This Level-1 DFD shows how data flows between external entities, processes, and data stores.

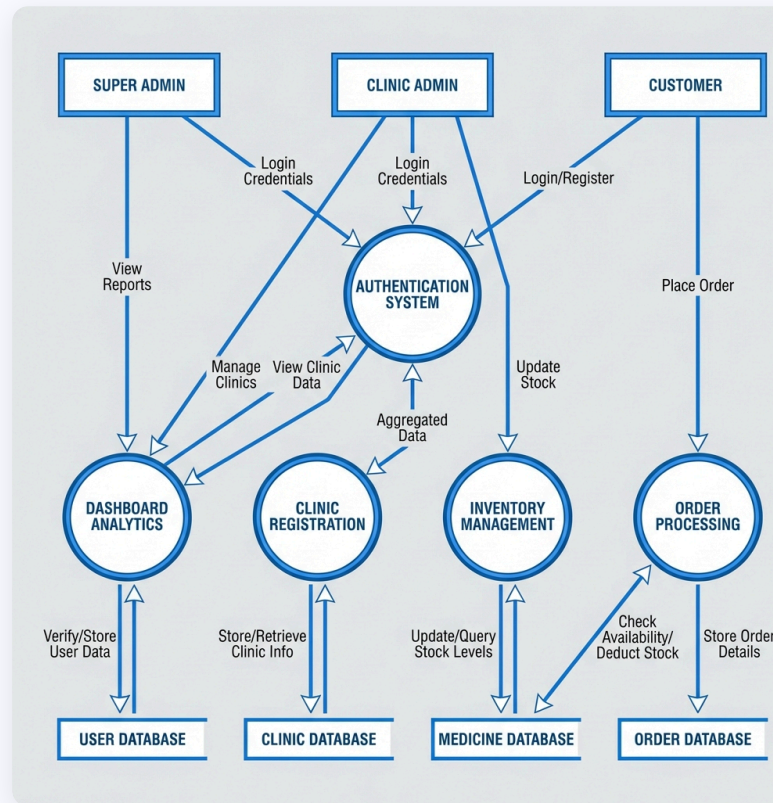


Figure 6: Level-1 Data Flow Diagram

External Entities

- **Super Admin** - Manages platform
- **Clinic Admin** - Manages pharmacy
- **Customer** - Places orders

Data Stores

- User Database
- Clinic Database
- Medicine Database
- Order Database

Sequence Diagram - Login Flow

Admin Authentication & Dashboard Access

This UML Sequence Diagram shows the complete login and data retrieval process.

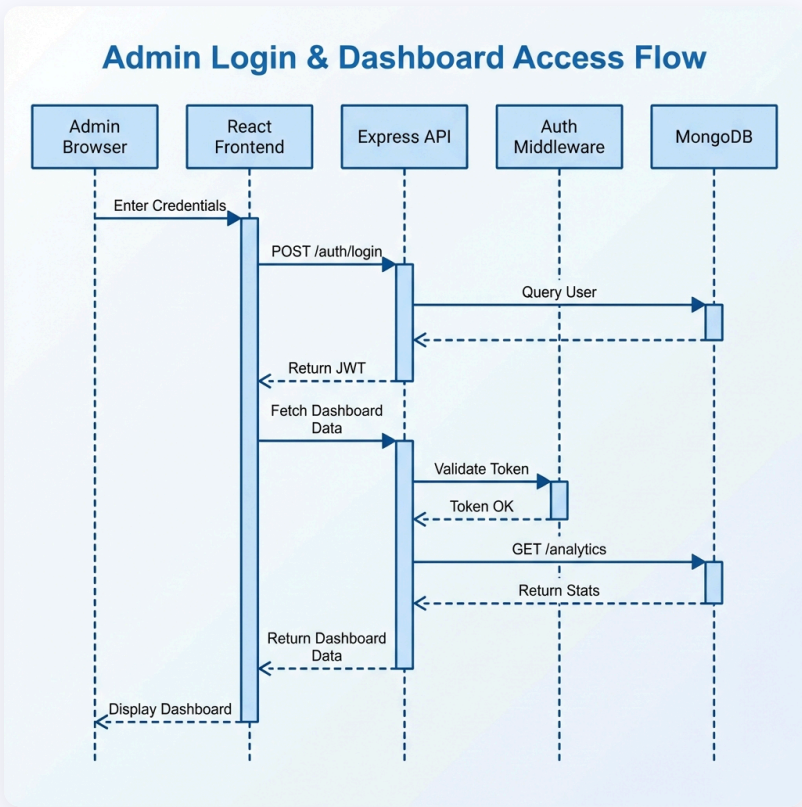


Figure 7: UML Sequence Diagram - Login Flow

Interaction Steps

#	From	To	Action
1	Admin	Browser	Enter credentials
2	Browser	Express API	POST /auth/login
3	Express	MongoDB	Query user
4	Express	Browser	Return JWT token

#	From	To	Action
5	Browser	Express API	GET /analytics
6	Express	Browser	Return dashboard data

State Diagram - Clinic Lifecycle

Clinic Status Transitions

This UML State Diagram shows all possible states a clinic can be in and the transitions between them.

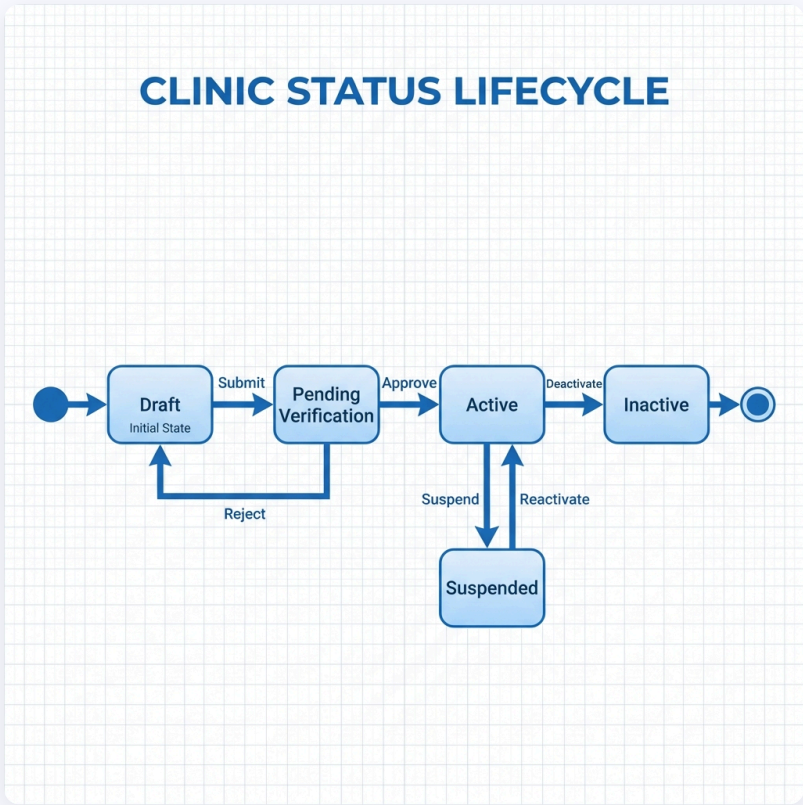


Figure 8: UML State Diagram - Clinic Lifecycle

State Definitions

State	Description	Allowed Actions
Draft	Incomplete enrollment	Edit, Submit, Delete
Pending Verification	Awaiting approval	Approve, Reject
Active	Fully operational	Suspend, Deactivate
Suspended	Temporarily disabled	Reactivate

State	Description	Allowed Actions
Inactive	Permanently closed	Archive

Dashboard UI Mockup

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Admin Interface Design

This mockup represents the Super Admin Dashboard interface with all key components.

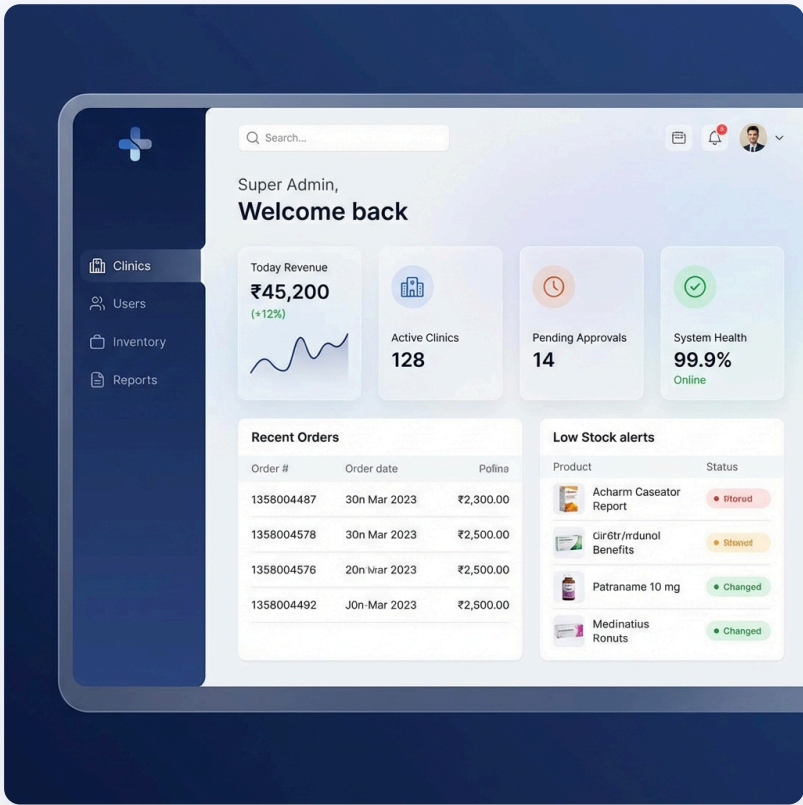


Figure 9: Admin Dashboard UI Mockup

UI Components

Component	Purpose
Sidebar	Navigation to Clinics, Users, Inventory, Reports
KPI Cards	Today's Revenue, Active Clinics, Pending Approvals, System Health
Recent Orders Table	Latest 5 orders with status

Component	Purpose
Low Stock Alerts	Medicines below threshold

Technical Recommendations

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Security Improvements

Recommendation	Implementation	Priority
Multi-Factor Authentication	Add OTP via SMS/Email for admin login	HIGH
Session Management	Implement single-session per admin	HIGH
Rate Limiting	Limit login attempts	MEDIUM

Performance Optimizations

Recommendation	Implementation	Priority
Redis Caching	Cache dashboard stats (5-min TTL)	HIGH
Database Indexing	Add indexes on frequently queried fields	MEDIUM
Lazy Loading	Load dashboard cards asynchronously	LOW

Feature Additions

Predictive Analytics

AI-based low stock forecasting

PDF Reports

Generate downloadable clinic reports

API Endpoints Summary

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Authentication Routes

Method	Endpoint	Description
POST	/api/auth/login	User login
POST	/api/auth/logout	User logout
GET	/api/auth/me	Get current user

Clinic Routes

Method	Endpoint	Description
GET	/api/clinics	List all clinics
POST	/api/clinics	Create new clinic
GET	/api/clinics/:id	Get clinic by ID
PUT	/api/clinics/:id	Update clinic
PUT	/api/clinics/:id/status	Update clinic status

User Routes

Method	Endpoint	Description
GET	/api/users	List all users
POST	/api/users	Create new user
PUT	/api/users/:id	Update user
DELETE	/api/users/:id	Delete user

Conclusion

Summary

This documentation provides a complete view of the **PharmaCare Admin Dashboard** system.

Document Highlights

- 8 Professional UML Diagrams covering all aspects
- Complete Data Model Documentation
- Step-by-Step Process Flows
- API Endpoint Reference
- Strategic Recommendations

Key Files Referenced

File	Purpose
Dashboard.jsx	Main admin dashboard component
ClinicList.jsx	Clinic management interface
UserList.jsx	User management interface
Clinic.js	Mongoose clinic model schema
User.js	Mongoose user model schema